

```

65.     secretNum = ''
66.     for i in range(NUM_DIGITS):
67.         secretNum += str(numbers[i])
68.     return secretNum
69.
70.
71. def getClues(guess, secretNum):
72.     """Returns a string with the pico, fermi, bagels clues for a guess
73.     and secret number pair."""
74.     if guess == secretNum:
75.         return 'You got it!'
76.
77.     clues = []
78.
79.     for i in range(len(guess)):
80.         if guess[i] == secretNum[i]:
81.             # A correct digit is in the correct place.
82.             clues.append('Fermi')
83.         elif guess[i] in secretNum:
84.             # A correct digit is in the incorrect place.
85.             clues.append('Pico')
86.     if len(clues) == 0:
87.         return 'Bagels' # There are no correct digits at all.
88.     else:
89.         # Sort the clues into alphabetical order so their original order
90.         # doesn't give information away.
91.         clues.sort()
92.         # Make a single string from the list of string clues.
93.         return ' '.join(clues)
94.
95.
96. # If the program is run (instead of imported), run the game:
97. if __name__ == '__main__':
98.     main()

```

After entering the source code and running it a few times, try making experimental changes to it. The comments marked with (!) have suggestions for small changes you can make. On your own, you can also try to figure out how to do the following:

- Change the number of digits for the secret number by changing `NUM_DIGITS`.
- Change the number of guesses the player gets by changing `MAX_GUESSES`.
- Try to create a version with letters as well as digits in the secret number.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens when you change the `NUM_DIGITS` constant?